From Episodes to Abstractions

Latent Hierarchical Memory in 1,908 AI Conversations

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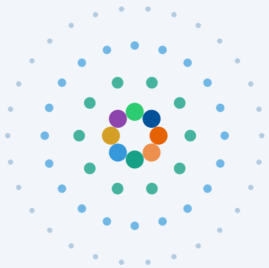

- outer: 1,908 episodes
- 500 meta-concepts
- 50 themes
- inner: 8 domains

github.com/queelius/chatgpt-complex-net

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Slide 1 – Title (35 s, ∼85w)

Good afternoon. I'm Alex Towell. Joint work with John Matta at SIUE.

In the last three years, millions of people have had thousands of conversations with LLMs. Each one is a trace of an idea being explored. What if we treated these archives not as chat logs, but as externalised semantic memory?

That's the question this paper asks. I'll show that one user's 1,908 ChatGPT conversations contain the same kind of structure cognitive scientists find in human semantic memory.

[Advance.]

The Problem: Archives Are Flat. Knowledge Is Not.

What you have

Chronological list

- Bayesian inference
- Cooking pasta
- Solomonoff induction
- R package debugging
- Philosophy of mind
- ⋮

Structure is invisible.

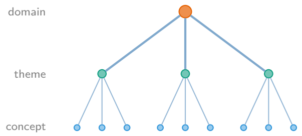
extract → concepts

The Question

Does **structure** emerge from a long archive of AI conversations — the way it does in human memory?

What's hidden

Latent hierarchy



If it's there, we should find it.

Slide 2 – The Problem (50 s, ∼115w)

[Gesture: left panel.] When you open your ChatGPT history, you see this: a chronological list. Bayesian inference next to cooking pasta next to Solomonoff induction. Adjacency in *time* tells you nothing about which conversations are about the same *thing*.

[Gesture: right panel.] But you know structure is there. You came back to Solomonoff three times last month; Bayesian inference is part of a domain you've been exploring. The hierarchy on the right is real — it's just invisible from a list.

So: can we recover that structure automatically? And is it the kind of structure cognitive science predicts for semantic memory?

To answer that, I borrow a framework from a different field.

Two Ideas from Cognitive Science

Hippocampal
fast, episodic
specific events

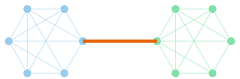
consolidate ⟷

Neocortical
slow, statistical
extracts regularities

Prediction:
New conversations reuse *existing* concepts more and more — vocabulary growth **slows**

[McClelland et al. 1995; Kumaran et al. 2016]

And: human semantic memory is a small-world network



Dense local clusters + a few global shortcuts.
Roget's, WordNet, word associations:
small-worldness ≈ 5–15

[Steyvers & Tenenbaum 2005]

The Hypothesis

If AI conversation archives are an *externalised* semantic memory, we should see both signatures: **sublinear vocabulary growth** *and* **small-world topology**.

Slide 3 – Cognitive Framework (1:15, ∼130w)

Two ideas from cognitive science.

[Gesture: left.] Complementary Learning Systems. The hippocampus encodes individual experiences fast; the neocortex extracts statistical regularities across many of them. As you learn, new experiences increasingly map onto categories you already have. We can measure that saturation as sublinear vocabulary growth.

[Gesture: right.] The structural side. Steyvers and Tenenbaum mapped human semantic memory as a network and consistently found small-world topology — dense local clusters plus a few long-range shortcuts. Small-worldness scores between roughly 5 and 15.

[Gesture: bottom block.] The hypothesis: if conversation archives are an externalised semantic memory, we should see *both* signatures. In one user's 1,908-conversation archive.

Here's how we tested it.

Pipeline: Concepts → Embeddings → Hierarchy

Raw episodes
1,908 conversations
Dec 2022 – Apr 2025

→

Extract concepts
LLM (Claude)
3–10 per episode

→

Embed
each concept
→ 768-d vector

→

Cluster
hierarchical, 4-level
episode → . . . → domain

Key instruction in the prompt: *"Be specific. Say 'Metropolis–Hastings algorithm,' not 'statistics.'"*

Output: a graph linking each episode to its concepts, then grouped into a four-level hierarchy.

Slide 4 – Pipeline (1:00, ∼120w)

[Gesture along the pipeline arrows.] Four stages.

Raw episodes: 1,908 ChatGPT conversations over 29 months from one user — me. Single-user case study; I'll come back to that.

Extract concepts: an LLM pulls three to ten concepts from each episode. *[Gesture: prompt box.]* The key detail is specificity. Not "statistics"; "Metropolis-Hastings algorithm." Specific concepts are what makes the geometry work later.

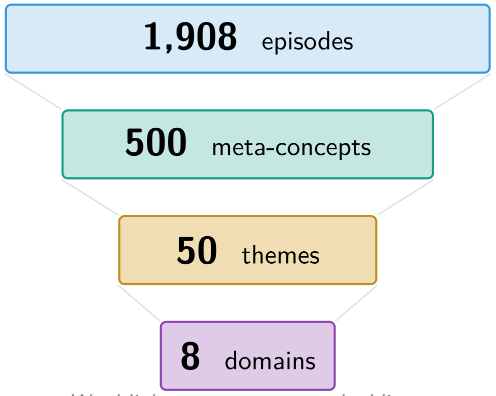
Embed: each concept becomes a 768-dimensional vector.

Cluster: hierarchical clustering, cut at three heights.

[Gesture: bottom output.] The output is a graph linking episodes to concepts, plus a four-level hierarchy. Everything that follows is just *measuring* this structure.

Here's what came out.

The Reveal: A Four-Level Hierarchy Emerges



Ward linkage on concept embeddings, cut at 3 heights · ∼3.5× compression per level

Eight interpretable domains

Software Engineering	1,074
Math & Optimisation	634
Statistical Methods	510
Philosophy & AI Theory	489
ML & Network Science	274
DevOps & Integration	273
AI Safety & Research	177
LLM Engineering	86

Two surprises

- Domains are **easy to name** — but there's no clean number of clusters.
- Most episodes **belong to several domains at once**.

Slide 5 – The Reveal (1:10, ∼145w)

[Pause. Let the picture land.]

[Gesture top to bottom along the cascade.] 1,908 episodes consolidate into 500 meta-concepts, then 50 themes, then 8 domains. Each level groups roughly 3.5 of the level below — the geometric signature of a hierarchy.

[Gesture: domain table.] The eight domains are easy to name. I didn't pre-specify them. Software engineering largest, then math, statistics, philosophy of AI, ML and network science, devops, AI safety, LLM engineering. They reflect my actual research and project history.

[Gesture: "Two surprises" block.] Two surprises.

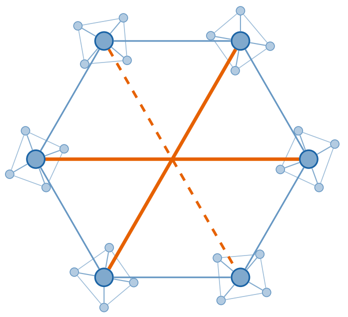
First: there's no clean number of clusters. Knowledge is a continuum, not a set of crisp categories. Eight domains is a useful summary, not a "right answer."

Second: most episodes belong to several domains at once. We'll come back to that.

Cognitive science predicts two signatures here. Both have to land for the analogy to hold. Here's the first.

Signature 1: Small-World Topology

Concept co-occurrence network
499 nodes, 5,935 edges



Compared to a random graph

[Humphries & Gurney 2008]

Local clustering	**6.89**×
Shortest paths	1.05×
Small-worldness	**6.6**

Matches human memory

(small-worldness)

Roget's Thesaurus	≈ 13
WordNet	≈ 15
Word associations	≈ 5.6
Our archive	**≈ 6.6**

[comparison values: Steyvers & Tenenbaum 2005]

Slide 6 – Small-World Topology (1:15, ∼130w)

[Gesture: network cartoon.] The concept co-occurrence network: 499 meta-concepts as nodes, edges when concepts share an episode.

[Gesture: clusters, then orange shortcuts.] Dense local clusters with a few long-range shortcuts. The Humphries-Gurney test compares clustering and path length against random graphs at matched size.

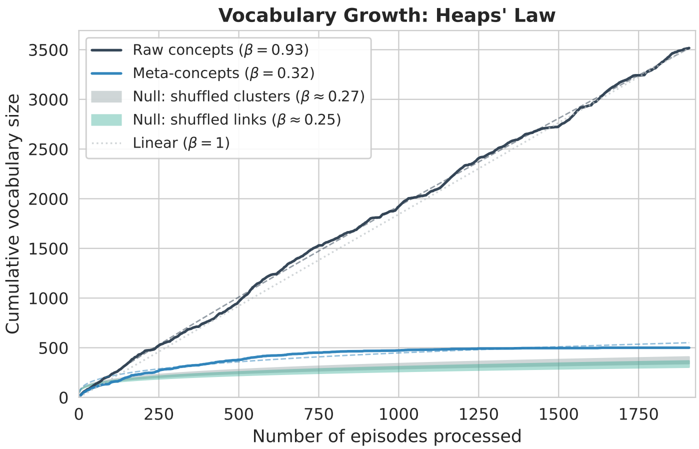
[Gesture: ratio table.] Clustering 6.9 times higher than random. Path length essentially unchanged. Small-worldness of 6.6.

[Gesture: human-memory block.] Roget's, WordNet, free-association norms give small-worldness between 5 and 15. The archive sits inside that range.

One caveat: I'm not claiming archives *are* semantic memory. They show the same structural signature — an externalised analog.

Signature one, confirmed. Now signature two: does the meta-concept layer consolidate the way CLS predicts?

Signature 2: Sublinear Vocabulary Growth



Heaps' law: $V(n) = K \cdot n^{\beta}$

[Heaps 1978]

Raw concepts	$\beta = 0.93$
Meta-concepts	$\boldsymbol{\beta = 0.32}$

Caveat: with k fixed at 500

Some saturation is automatic. So we compare against **null models**.

Two nulls, both reject

- Shuffled clusters: $\beta = 0.27$
- Shuffled links: $\beta = 0.25$
- Both $p < 10^{-3}$

Slide 7 – Vocabulary saturates (1:15, ∼135w)

[Gesture: plot.] Heaps' law: vocabulary grows like n^{β}. Real text has β around 0.4 to 0.6. Lower means stronger saturation — the consolidation CLS predicts.

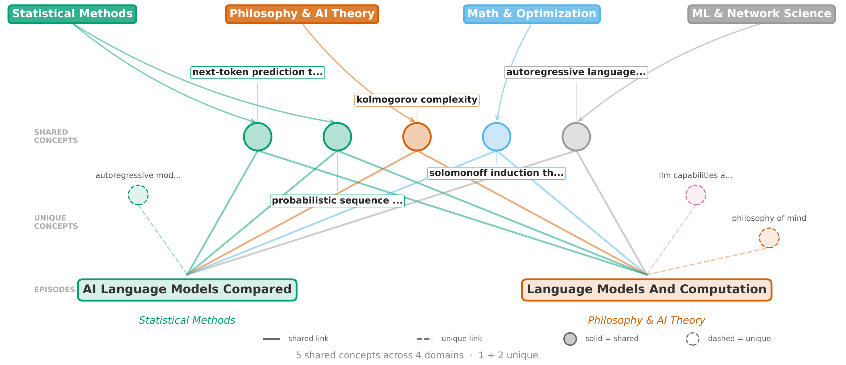
[Gesture: top curve.] Raw concepts grow nearly linearly, $\beta = 0.93$. We keep inventing new noun phrases. *[Gesture: bottom curve.]* Meta-concepts: $\beta = 0.32$. Strongly sublinear.

[Gesture: caveat.] But fixing the number of clusters at 500 means some saturation is automatic. We need null models.

[Gesture: two-nulls block.] First null: shuffle which raw concept goes in which cluster. $\beta = 0.27$. Second null: shuffle the episode-concept links while keeping the row and column totals. $\beta = 0.25$. Our 0.32 exceeds both at $p < 10^{-3}$.

Why *higher*, not lower? Coherent clusters draw real distinctions; random clusters absorb anything. The saturation is real — and it only showed up because we kept the set-based structure. Which is the next slide.

77% of Episodes Span Multiple Domains



One-label methods *(Louvain, k-means, BERTopic)*: each episode gets one cluster. These two end up apart — connection vanishes.

Set-based methods: each episode is a *set* of concepts. Two episodes share concepts across *four* domains and stay linked.

Slide 8 – 77% span multiple domains (1:10, ∼125w)

[Gesture: the two episode boxes.] Two real episodes. One lives in Statistical Methods. The other lives in Philosophy and AI Theory. Different domains.

[Gesture: the shared concept list in the middle.] But they share five concepts — Kolmogorov complexity, Solomonoff induction, and three others — and those five concepts span *four* different domains.

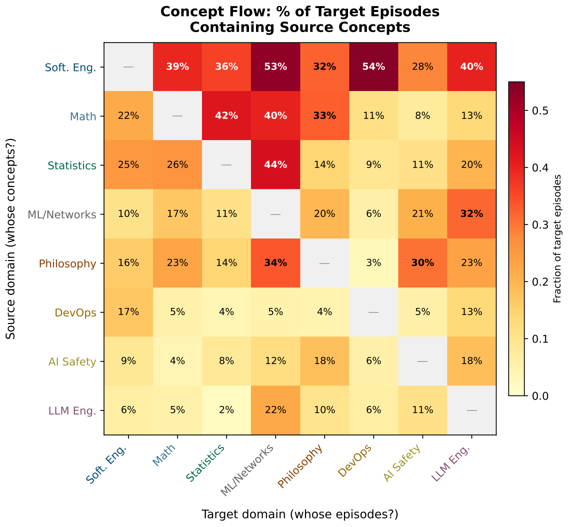
Any one-label-per-episode method assigns each episode to a single cluster. These two episodes end up in different clusters and the connection *vanishes*.

The set-based model preserves it. An episode *is* a set of concepts; two episodes link through any shared concept, regardless of which domain that concept lives in.

And this isn't rare: **77% of episodes span two or more domains; 35% span three or more.** Cross-domain is the rule. That's the methodological contribution.

With this structure, we can ask a question single-label methods can't.

Who Borrows From Whom? (Concept Flow)



Two questions per domain

Where do its concepts go? (rows)

Which concepts come in? (columns)

What the heatmap says

- On average **39% below** random
  ⇒ domains really are separate
- LLM Eng. → ML/Net: **2.67**× random
  ⇒ specific dependencies

Software Eng. broadcasts to all 7 others.

Slide 9 – Who borrows from whom (1:10, ∼135w)

[Gesture: heatmap.] For each pair of domains (A, B): what fraction of B's episodes contain a concept from A? Reading across a row tells you where A's concepts go. The reverse direction is usually different — asymmetry an undirected network can't see.

[Trace Software Engineering's row.] Software Engineering broadcasts into every other domain. *[Gesture: math↔statistics.]* Math and statistics feed each other heavily. *[Gesture: LLM Eng. row.]* LLM Engineering is small but punches above its weight.

The catch: Software Engineering owns 30% of raw concepts. Are we just measuring size? We control against a null that shuffles concept–domain labels while keeping the sizes fixed.

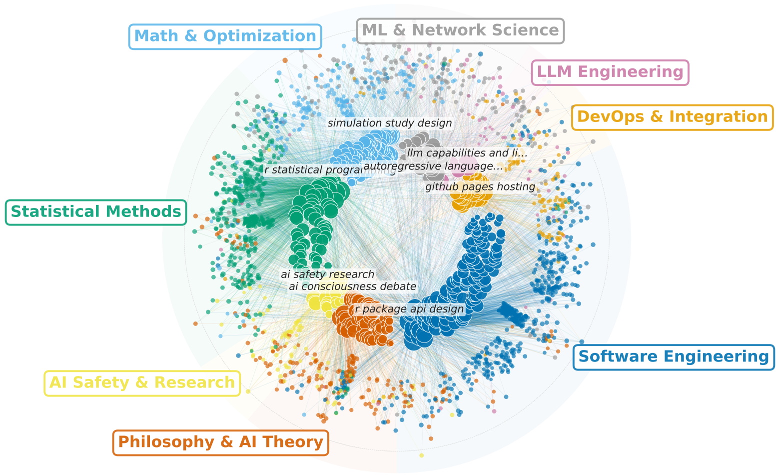
On average, sharing comes in **39% below random**. Domains really are separate.

[Gesture: heatmap-says block.] But specific pipelines stand out. LLM Engineering → ML and Networks: **2.67**× the random expectation. That's a targeted dependency, not spillover.

The story: *real boundaries, with a few specific pipelines.*

Here's what the whole thing looks like.

The Full Picture



Inner: 499 concepts (coloured by domain). Outer: 1,908 episodes. Edges: episode→concept.

Slide 10 – The Full Picture (35 s, ∼70w)

[Gesture: the whole image; caption already names the rings.] The whole archive in one image.

[Trace Software Engineering's sector.] Software Engineering sends links everywhere. *[Gesture to Philosophy.]* Philosophy and AI Theory stays mostly within its own sector. Statistics, math, AI safety, the small LLM sliver — each visible as a distinct coloured region.

[Pause for the gestalt.]

Three takeaways.

Takeaways & What's Next

What we found

- 4-level hierarchy
  1,908 → 500 → 50 → 8
- Small-world structure matches human memory
- Vocabulary saturates as CLS predicts
- 77% span $\geq$ 2 domains

Why it matters

- Archives = **externalised semantic memory**
- Natural index for **concept-level Graph RAG**
- Knowledge is a **continuum, not a partition**

Limits & next

- **N=1** single user. Replicate on WildChat
- No natural k
- Extend to **agentic** sessions

Treat AI conversation archives as **structured cognitive artifacts** —
the structure was there all along; we just needed the right lens.

Questions? queelius/chatgpt-complex-net

Slide 11 – Takeaways (1:10, ∼155w)

Three angles to take away.

[Gesture: "What we found" column.] Four-level hierarchy: 1,908 episodes, 500 concepts, 50 themes, 8 domains. Small-world structure matching human-memory benchmarks. Vocabulary saturates, as CLS predicts. And most episodes span multiple domains.

[Gesture: "Why it matters" column.] A principled way to treat AI archives as externalised semantic memory. A natural index for concept-level retrieval over your own conversations. And: knowledge is a continuum, not a partition.

[Gesture: "Limits and next" column.] N equals one — next step is replicating on WildChat. No natural number of clusters. And we're extending the method to *agentic* sessions, where parent-child delegation adds a layer.

[Gesture: bottom takeaway box.] The headline: treat AI conversation archives as structured cognitive artifacts. The structure was there all along; we just needed the right lens.

Thank you. Code and data at the link. Happy to take questions.

Backup: Method Details

Concept extraction

- Model: `claude-3-5-sonnet-20241022`, T=0
- Per episode: first 20 messages, 500-char cap
- Target: 3–10 noun phrases, 2–5 words
- 20 parallel agents (no shared vocabulary)
- Post-hoc dedup: geometric (no string matching)

Embedding

- Model: `nomic-embed-text`
- 768-dim, 8k context, L2-normalised

Hierarchical clustering

- Ward linkage, Euclidean distance
- Cuts at $k = 500, 50, 8$
- Silhouette monotonic for $k \geq 20$ (no natural boundaries)

Yield

- 7,555 raw concept mentions
- 3,517 unique (after case dedup)
- 500 meta-concepts after Ward
- 7,153 bipartite edges
- 5,935 co-occurrence edges on 499 nodes

Backup: Robustness to k

k	Heaps' β	Small-worldness	Clustering C	Edges
50	0.061	1.1	0.830	906
100	0.103	1.4	0.652	2,286
200	0.158	2.0	0.425	4,111
300	0.225	2.9	0.351	5,073
500	**0.320**	**6.57**	**0.329**	**5,935**
700	0.382	12.6	0.351	6,593
1000	0.467	25.9	0.401	7,291

Qualitative conclusions (small-worldness > 1, sublinear β, null rejection) hold for every k tested.
Quantitative numbers shift smoothly: the 500-cluster cut is a useful summary, not a privileged scale.